AI Model

Input → Process → Output

AI Agent

A system that uses AI to achieve a goal by deciding what to do and take actions.

Example: Imagine an AI agent that detects fake reviews one e-commerce platform.

1. User provides the product URL.
2. AI agent decides what to do
    a. Fetch review.
    b. Predict the review is positive or negative.
    c. Generate the report.

In short Model make prediction and Agent manages overall task.

Example: AI travel Agent

We can tell AI to plan a trip form Coimbatore to Chennai within 5000k.

AI agent will understand the goal, search for travel options, accommodation and prices, and then suggest a suitable travel plan.

Google ADK (Agent Development Kit)

A Framework developed by a google, that helps developers to build, test and deploy AI Agents.

Chat-GPT → It is a ready-made application, it helps you to interact with AI.

AI Agent → It helps you to build your own AI Agents, so that it can perform task as per your requirement.

Example: In Chat-GPT you paste the review and it will answer whether the review is positive or negative.

But with the help of an AI agent, you connect it to the database or a model and then you can get your answer.

➔ In the AI Agent your data never leaves you environment.

Agent Card

An Agent Card is like an agent's introduction or business card. It tells other applications or agents what an agent can do and how to contact it.

For example, your time agent might advertise:

1. Name: Time Agent

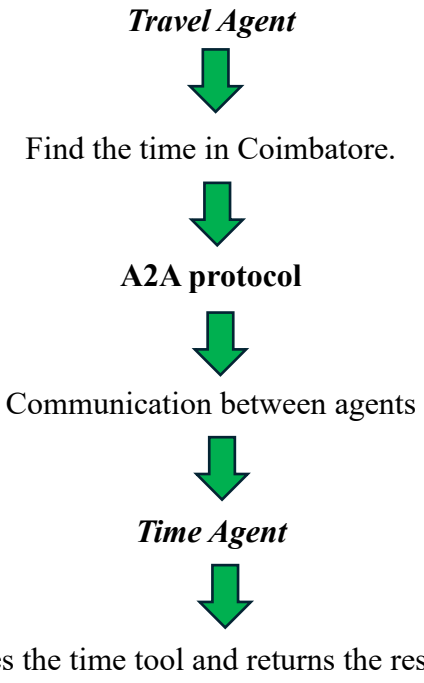
2. Purpose: Answer questions about the time in a city.
3. Skills: Get the current time.
4. Endpoint: The address where the agent can be contacted.
5. Capabilities: The communication features it supports.

An Agent Card is commonly used in the A2A (Agent2Agent) protocol so other agents can discover and understand your agent.

Agent 2 Agent (A2A)

It is an open protocol that allows separately built AI agents to communicate and collaborate.

Imagine you have two agents:



The Travel Agent does not need to know the Time Agent's internal Python code. It can communicate through the agreed protocol.

A2A Client

The part that sends a request to another agent.

Example: The Travel Agent asks the Time Agent for the current time.

A2A Server

It receives the request form the other agent and returns the response.

Example: Travel Agent sends request to the Time agent.

Session

Maintain Conversation or it keeps the track between you and AI.

Example:

- You: What is the time in Coimbatore?
- Agent: 10:30 AM.
- You: What about Chennai?

State

Stores the specific information of a particular session. It is temporary.

Example: You tell AI that my favourite city is Chennai.

It will only store this information in this chat, if you try to open another chat or new chat and tell what is my favourite city, it will say I don't know.

Memory

Make selected information available for future sessions.

Example: You tell AI, store my favourite city as Chennai in the memory.

So now when you open the new chat, and then tell AI what is my favourite city, it will respond you Chennai.

Chat GPT is an AI Agent but the model it uses behind is OpenAI.

Ollama → Software that runs your AI models like **llama 3.1** on computer. Your application send prompt it responds.

In the 2nd Module Lab Part:

First Run all the cells up to step 5. Then,

Go to Files → New Terminal (Open 3 terminals)

In 1st terminal copy the below command and wait for it to run completely:

```
vllm serve meta-llama/Llama-3.1-8B-Instruct \
  --enable-auto-tool-choice \
  --tool-call-parser llama3_json \
  --port 8088 \
  --chat-template vllm/examples/tool_chat_template_llama3.1_json.jinja \
  --gpu-memory-utilization 0.6
```

In the 2nd terminal copy the below command and wait for it to run completely:

```
curl http://localhost:8088/v1/models
```

In the 3rd terminal copy the below command and wait for it to run completely:

1. nohup ollama serve > /tmp/ollama.log 2>&1 &
2. ollama run llama3.1
3. curl http://localhost:11434/api/tags

First paste the step 1 command and run it and then step 2 and then step 3.